university of sussex • 21 october 2026

# Physics 2.0

## Formalisation and Physlib: a synergy between AI, mathematics and physics

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github.com/leanprover-community/physlib josephtoobysmith.com

section 1 of 6

# The problem we face

**The problem**
Truth
Trust
Community
Recognition
Conclusion

# The problem we face

1. AI is getting better and better.
2. We don’t know how far it will get.
3. The arXiv and journals are being overwhelmed.
4. It is hard to know what to trust and what not to.

# Evidence that AI is getting better

**july 2026**

## The Jacobian conjecture

Keller, 1939. Counterexample in $\mathbb{C}^3$.
Alpoge, with Claude Fable 5

**august 2026**

## Ten open problems

From maths and CS
OpenAI “Astra”

**september 2026**

## Navier–Stokes problem

Finite time blow-up.
AI models

**october 2026**

## 372 problems

OpenAI, internal model

## Things to preserve: four tenets of academia

**Truth**

**Trust**

**Community**

**Recognition**

# Physics 1.0

**truth**

Disseminated in academic papers.

**trust**

Peer review.

**community**

Spends its time writing papers, and reviewing them.

**recognition**

Citation counts, and numbers of papers.

# Do not fear: there is a solution

I will call it **Physics 2.0**.

# Do not fear: there is a solution

I will call it **Physics 2.0**.

AI, while pushing us away from papers, has the counterpoint of now making the move away from papers **possible**.

section 2 of 6

# Reorganising truth

The problem

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## What is wrong with papers?

1. Hard to find information.
2. Hard to reuse.
3. Hard to find connections.
4. Papers are outdated. It is time to find something new.

# The solution

Store information not in papers, but in an **open-source, community-run** corpus of knowledge.

Easy to **find** information.

Easy to **reuse**.

Easy to find **connections**.

Similar reason to why **SciPy** exists in Python.

# Dissent against our solution

1. Such a corpus of knowledge is hard to maintain.
2. It is hard to keep up to date.

AI helps with **all of this**.

# Ingredient 1 of Physics 2.0



**leanprover-community / Physlib**

A community-driven project to digitalise results from physics into Lean.

public

lean 4

★ **786** stars  **205** forks  **114** contributors  started 2024

I will explain the details later.

section 3 of 6

# Adjusting trust

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# What is wrong with peer review?

1. It does not scale efficiently with AI.
2. Mistakes get into papers, and slip past reviewers.

# The mathematicians have a solution

It is called **formalisation**.

# Technical interlude

# What is an interactive theorem prover?

A programming language in which you can write mathematical theorems proofs, and definitions, with mathematical **correctness** and **consistency** guaranteed. And the definitions you write are **executable**.

# What is formalisation?

The process of writing results into an **ITP**.

# The four main interactive theorem provers

**Rocq** **Isabelle/HOL** **Lean** **Agda**

This talk is in **Lean**.

# A rapid overview of Lean

## Proof compiler

Definitions, theorems, proofs
⟶ **consistency & correctness, guaranteed**

## Built on type theory

Lean knows the **type** of everything, and checks that things have the type expected.

$$\underbrace{p}_{\text{proof}} : \underbrace{P}_{\text{theorem}} \qquad \underbrace{7}_{\text{term}} : \underbrace{\mathbb{N}}_{\text{type}}$$

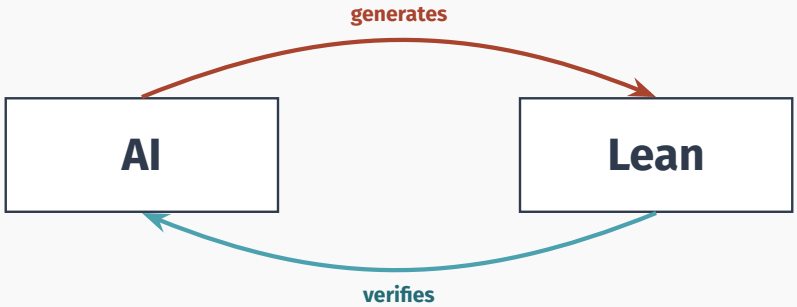
## An example

```
def energy (m v : Real) : Real :=
  m * v ^ 2 / 2

theorem energy_nonneg
    (m v : Real) (hm : 0 <= m) :
    0 <= energy m v := by
  unfold energy
  positivity
```

Open this example in the live editor

# The formalisation–AI loop



# End of technical interlude

# How good is AI-driven formalisation now?

It can essentially formalise **anything you want**.

Anthropic's Claude formalised Fermat's Last Theorem in Lean in about eleven days, on Lean's three standard axioms alone and with no `sorry`s.

# How good is AI-driven formalisation now?

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However, this does not necessarily mean **well**.

# Formalisation does not give you complete trust

Formalisation is not the whole picture, and should not give you the whole trust.

## Fermat's Last Theorem, made trivial

```
-- "the integers"
def MyInt : Type := Empty

def add (a b : MyInt) : MyInt :=
  Empty.elim a
def pow (a : MyInt) (n : Nat) : MyInt :=
  Empty.elim a

theorem flt (a b c : MyInt)
    (n : Nat) (hn : 3 <= n) :
    Ne (add (pow a n) (pow b n)) (pow c n) :=
  Empty.elim a
```

Lean accepts it. Nothing here is about numbers.

# Ingredient 2a of Physics 2.0

**Physlib** is a **corpus of knowledge** whose content is **formalised into Lean**.

Community run, and open source.

**a dual process**

## Lean verification

the mathematics

**and**

## Human review

the physics context, style, organisation and quality

# The problem

Lean verification takes the mathematics. But part of the trust still rests on **human review** — and that **cannot scale**.

# Ingredient 2b of Physics 2.0

There is another solution to scaling trust: **one-look review**, similar to the arXiv.

**PhyslibAlpha**

This is **PhyslibAlpha**, and it allows things to **scale**.

# Project structure

upstream **Mathlib**

core

## Physlib

Curated to a high standard for long-term reuse.

rapid

## PhyslibAlpha

Lighter review, for large-scale AI-assisted contributions.

qi

## QuantumInfo

Quantum information theory, being brought closer to Physlib.

adjacent **CSLib**

# Physlib is one of three main ones



leanprover-community /

## mathlib4

The math library of Lean 4.

★ 4,309 ⑂ 1,768
👥 798 🕒 2021



leanprover-community /

## Physlib

Physics, digitalised into Lean. Community-driven.

★ 786 ⑂ 205
👥 114 🕒 2024



leanprover /

## CSLib

The computer science library of Lean.

★ 754 ⑂ 214
👥 50 🕒 2025

# Evidence that this can be done at scale and works

Top-level directories of `Physlib/`, by lines of Lean.

| | | | |
|---|---|---|---|
| Relativity | 27,545 | StatisticalMechanics | 2,449 |
| QFT | 20,162 | ProbabilisticTheory | 2,306 |
| Particles | 19,104 | Cosmology | 1,858 |
| SpaceAndTime | 16,069 | Meta | 1,773 |
| Mathematics | 15,795 | FluidDynamics | 1,361 |
| QuantumMechanics | 13,057 | CondensedMatter | 1,003 |
| ClassicalMechanics | 11,784 | ClassicalFieldTheory | 554 |
| Electromagnetism | 9,442 | Thermodynamics | 432 |
| StringTheory | 5,285 | Optics | 48 |
| Units | 4,076 | LatticeQFT | 22 |

**154,125** lines of Lean across **619** files.

## Evidence that this can be done at scale and works

# The Higgs boson, in Lean

live.lean-lang.org

# Additional benefits of having Physlib

1. Allows verification of results, both human and AI.
2. Allows easy reuse, and improvement.
3. Allows assumptions to be tracked.
4. Allows one to easily explore and constrain model spaces.
5. Prevents bit-rot: one corpus, kept compiling.
6. Avoids duplicated work, and definitions that are incompatible with one another.
7. AI speed and efficiency makes this feasible.

section 4 of 6

# Community

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## What the community currently does

# Writes papers.
# Reviews for journals.

# What will they do in this new world?

Writing papers, and reviewing for journals, is exactly the work that AI is now **capable of**.

# Ingredient 3 of Physics 2.0

Working on the corpus of knowledge — **Physlib**.

- Review
- Maintenance
- Refactoring
- Foundational work
- Formalisation
- Documentation
- Extending
- Adding experiments
- Building tools

# People are already doing this

**100+**
contributors

**7**
maintainers

**Always looking for more**

# Built for the community

**we refactor**

When a better approach is found, we take it.

**a choice of starting point**

There is no single correct axiomatisation, so we allow a **choice of starting point**.

# Difficulties in formalising physics

1. Validation does not come with the guarantees of verification.
2. One physical concept, many mathematical descriptions.
3. Physics papers aren't written as definitions and theorems.
4. How physical assumptions enter.

section 5 of 6

# Recognition

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# The problem with current recognition

Recognition is currently based only on **publishing papers** and **citations**.

In this new world that is a **poor proxy** for the work that actually matters — because it is precisely the work we have seen AI is **very capable of**.

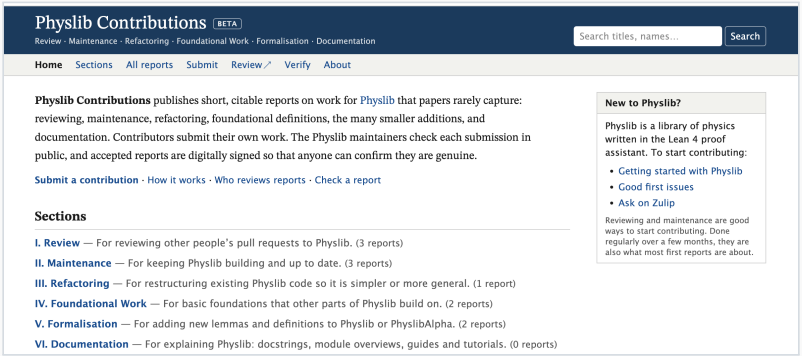
# Who gets the recognition?

The library is the **foundation of truth**.

A separate, **arXiv-like system** records contributions — **verified**, and usable on a CV.

# Ingredient 4 of Physics 2.0

## contributions.physlib.io



section 6 of 6

# Conclusion

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# Physics 2.0

**truth**

Papers → **Physlib**

**trust**

Peer review → **Lean + human review**

**community**

Writing papers → **Building and maintaining the corpus**

**recognition**

Citations → **Contributions recorded and verified**

# Call to action

**if you are hiring, or on grant committees**

Start to recognise contributions to projects like **Physlib**, **Mathlib** and **CSLib** on a par with papers.

**if you are an expert in physics**

Come to **Physlib**, help review, help push forward your area of expertise.

# Thank you for watching

**Joseph Tooby-Smith**
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github.com/leanprover-community/physlib contributions.physlib.io josephtoobysmith.com